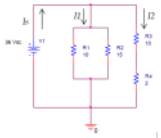
A1 Assynchronous Class Deliverable: Electronic Circuit Analysis Fundamentals

Saturday, August 29, 2026 8:11 AM

Figure 1.



(a) Calculate the total equivalent resistance of the circuit

$$R_t = (R_1||R_2)||(R_3 + R_4) = (10||15)||(10+2) = (\frac{1}{\frac{1}{10}+\frac{1}{15}})||(12) = \frac{1}{\frac{1}{6}+\frac{1}{12}} = 4\Omega$$

(b) Calculate the values of the currents Is, I1 and I2

$$I_1 = \frac{V1}{R1||R2} = \frac{36}{\frac{1}{\frac{1}{10}+\frac{1}{15}}} = \frac{36}{6} = 6A$$

$$I_2 = \frac{V1}{R3+R4} = \frac{36}{10+2} = \frac{36}{12} = 3A$$

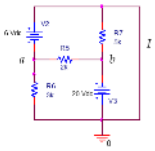
(c) Calculate the value of the voltage that appears across the 2 ohm resistor

$$\mathrm{V_{R4}} = I_2 * R4 = 3 * 2 = 6V$$

(d) Calculate the value of the current that passes through the 15 ohm resistor

$$\mathrm{I_{R2}} = \frac{V1}{R2} = \frac{36}{15} = 2.4V$$

Figure 2.



(a) The voltage of node a – ground and node b – ground

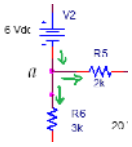
$$\mathrm{V_{a-ground}} = \mathrm{V_a} - \mathrm{V_{ground}} = -6 - 0 = -6V$$
$$\mathrm{V_{b-ground}} = \mathrm{V_b} - \mathrm{V_{ground}} = -20 - 0 = -20V$$

(b) Determine the voltage between nodes a – b, that is, Vab

$$\mathrm{V_{ab}} = \mathrm{V_a} - \mathrm{V_b} = -6 - (-20) = 14V$$

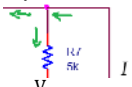
(c) Determine the value of the current I and indicate its positive direction.

Node a current analysis:



$$\mathrm{I_{V2}} = \mathrm{I_{R6}} + \mathrm{I_{R5}} = \frac{\mathrm{V_{a-ground}}}{R6} + \frac{\mathrm{V_{a-b}}}{R5} = \frac{-6}{3k} + \frac{14}{2k} = -2mA + 7mA = 5mA$$

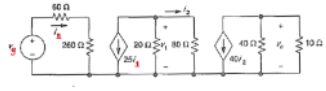
Top node current analysis:



$$\mathrm{I} = \frac{\mathrm{V_{ground-b}}}{R7} + I_{V2} = \frac{0-(-20)}{5k} + 5mA = \frac{20}{5k} + 5mA = 4mA + 5mA = 9mA$$

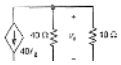
I current direction is from bottom to top

Figure 3.



Calculate the value of the voltages vg and V1 of the circuit in figure 3 if the voltage Vo is equal to 5 volts. Hint: perform the analysis from right to left.

Top right node current analysis:



$$40I_2 + \frac{V_0}{40\Omega} + \frac{V_0}{10\Omega} = 0A$$

$$Thus, I_2 = \frac{-\frac{5V}{40\Omega} - \frac{5V}{10\Omega}}{40} = \frac{-0.125A - 0.5A}{40} = -15.625mA$$

V1 calculation:

$$V_1 = I_2 * 80\Omega = -15.625mA * 80\Omega = -1.25V$$

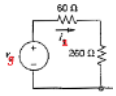
Top middle node current analysis:



$$25I_1 + I_{V1} + I_2 = 0A$$

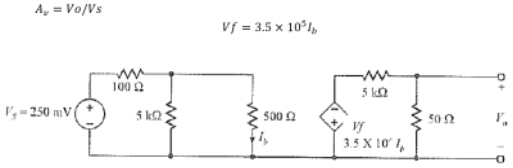
$$Thus, I_1 = \frac{-\frac{-1.25V}{20\Omega} - (-15.625mA)}{25} = \frac{62.5mA + 15.625mA}{25} = \frac{78.125mA}{25} = 3.125mA$$

Vg calculation:



$$V_g = I_1 * 60\Omega + I_1 * 260\Omega = 3.125mA * (60\Omega + 260\Omega) = 3.125mA * (320\Omega) = 1V$$

Figure 3. A typical transistor amplifier in its small signal model is shown.



Find the gain of the amplifier, that is, the ratio of the output voltage to the input voltage.

V_b calculation:

$$R_{5k\Omega||500\Omega} = \frac{1}{\frac{1}{5k\Omega} + \frac{1}{500\Omega}} = 454.54\Omega$$

$$V_b = \frac{V_s * 454.54\Omega}{100\Omega + 454.54\Omega} = V_s * 0.81$$

$$I_b = \frac{V_b}{500\Omega} = \frac{V_s * 0.81}{500\Omega} = V_s * \frac{1.62}{k\Omega}$$

V_0 calculation:

$$V_f = 350k\Omega * I_b = 350k\Omega * V_s * \frac{1.62}{k\Omega} = V_s * 567$$

$$V_0 = \frac{-V_f * 50\Omega}{5k\Omega + 50\Omega} = \frac{-V_s * 567 * 50\Omega}{5050\Omega} = \frac{-V_s * 28350}{5050} = -Vs * 5.61$$

$$thus, A_v = \frac{V_0}{V_s} = -5.61$$